

ZIQI CHEN

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EDUCATION

Rensselaer Polytechnics Institute, NY, United States

September 2022 - present

Ph.D.

School of Architecture

Architectural Acoustics program

Rensselaer Polytechnics Institute, NY, United States

September 2020 - August 2021

Master of Science

School of Architecture

Architectural Acoustics program

University of Rochester, NY, United States

September 2018 - August 2020

Master of Science

Department of Electrical and Computer Engineering

Suzhou University of Science and Technology, Jiangsu, China

September 2014 - July 2018

Bachelor of Engineering

Department of Electrical and Communication Engineering

FELLOWSHIP & SCHOLARSHIP

Humanities, Arts, Architecture, and Social Sciences Fellowship 2025-2026

(\$ 36000)

Rensselaer Polytechnic Institute (RPI) offers competitive fellowships for graduate students in Humanities, Arts, and Social Sciences (HASS) and Architecture, providing stipends and tuition support for promising PhD candidates.

INCE Student Scholarship 2023

(\$ 6000)

The INCE Student Scholarship is granted to a college student studying noise control or architectural acoustics in the United States of America who, through personal qualifications and proposed research experience, is judged to exhibit the highest potential for performing research benefiting some aspect of noise control and promoting its usefulness to society. An application must be received for each candidate wishing to be considered for this award.

AWARD

The Leo Beranek Student Medal 2025

The Leo Beranek Medal was established by the Board of Directors of INCE/USA on October 24, 2010 to recognize excellence in the study of Noise Control Engineering. It is awarded annually to outstanding undergraduate and graduate students at those academic institutions in North America that have courses in, or related to, Noise Control Engineering, including practical applications.

The Robert Bradford Newman Student Award 2021

The Robert Bradford Newman Student Medal for Merit in Architectural Acoustics recognizes excellence

in the study of acoustics and its application to architecture. The program honors outstanding students at schools of architecture and architectural engineering throughout the world. Students selected for the Newman Medals must have demonstrated excellence in this discipline and in the application of acoustical design principles in the course of their study.

PROJECTS

Acoustic measurement calibration based on MCMC methods

Use MCMC methods (like Nested sampling and Importance sampling) to estimate critical acoustical parameters accurately. I also developed an easy-to-use UI for further data collection. The accuracy of measurement results after calibration is improved by 50-100%.

Surface impedance model using Machine Learning

A novel approach is proposed for estimating the surface impedance and admittance of a wall, utilizing the Bayesian parameter estimation model selection method. This method enables the selection and estimation of parameters for any given model, even without precise knowledge of the underlying model mechanics. By employing this method, cost models can be simplified to a manageable complexity level while preserving accuracy.

Room acoustic simulation

A multipole model is introduced into the Finite-difference time-domain (FDTD) method. The arbitrary boundary conditions are expressed in the form of multipole functions. Auxiliary differential equation is used to reduce the computational load of FDTD method. The simulation is implemented for both 2D and 3D designs in Matlab. The parameters in the multipole model are estimated using the Bayesian inference based on the experimental measurement. The simulation results (acoustic reflectance and surface impedance) are close to the experimental measurement and the model.

Impedance tube design

Design the platform for the impedance tube measurement, including designing and modifying the tube with different materials, coding UI for impedance/admittance calculation, and platform configuration. The transfer function method employs an air layer with a rigid termination to accurately estimate microphone positions and dissipation. The measurement results are then predicted using finite-difference time-domain simulations. Real-time simulation is implemented in Matlab to forecast the acoustical performance of the porous material. The user interface incorporates the estimation of air dissipation, sound speed, and other general acoustical properties.

Architecture field measurement

Several acoustical buildings in New York state capital region are measured acoustically using an omnimicrophone, a figure-8 microphone, and a dummy head. The measured buildings include Troy Savings Bank Music Hall, Cohoes Music Hall, Proctors Theatre, Seiji Ozawa Hall, Universal Preservation Hall, Experimental Media and Performing Arts Center(EMPAC) at RPI, and Troy Gasholder House. The acoustical properties of these halls, such as reverberation time, T30, and early decay time, are calculated. The results are analyzed to find the potential problems in the building. Several treatment suggestions are proposed.

Flat-panel Loudspeaker Drivers Design

The objective of the project is to design flat-panel loudspeaker drivers for personal electronic devices. The shakers for small screens in personal devices are analyzed using the Finite Element Method (FEM). The mechanical and acoustic properties of the actuators are developed to accurately describe the bending motion of the thin plate. An array of actuators is utilized to control the bending modes of the flat panel effectively.

Underground Odor Source Localization Employing a Zigzag Searching Strategy

The search behavior of lower organisms is analyzed to develop search strategies that can be employed to locate underground odor sources, such as chemical contamination resulting from leakage in underground pipelines. An adaptive method is utilized to minimize costs and improve the efficiency of the source search.

JOURNAL PUBLICATIONS

1. **Z. Chen** and N. Xiang, "Bayesian estimation of dissipation and sound speed in tube measurements using a transfer-function model," *J. Acoust. Soc. Am.*, vol. 155, no.4, pp. 2646-2658, 2024.
2. N. Xiang, M. Hoeft, C. J. Fackler, **Z. Chen**, and P. Barach, "Validation of Bayesian design for broadband microslit panel absorbers in multiple layers," *J. Acoust. Soc. Am.*, vol. 154, no. 3, pp. 1471-1481, 2023.
3. **Z. Chen**, N. Xiang, and K. V. Horoshenkov, "Boundary admittance estimation for wave-based acoustic simulations using Bayesian inference," *J. Acoust. Soc. Am. Express Lett.*, vol. 2, no. 8, pp. 081601 (1-7), 2022.
4. **Z. Chen**, N. Xiang, and C. J. Fackler, "Bayesian estimations of dissipation, sound speed, and microphone positions in impedance tubes," *J. Acoust. Soc. Am. Express Lett.*, vol. 2, no. 8, 083601 (1-6), 2022.
5. J. Wang, **Z. Chen**, Z. Lei, "Olfaction Threshold-Based Thought of Micro-Steps Accumulation and Its Application on Odor Source Search," *IEEE Sensors J.*, vol. 18, no. 17, pp. 7294-7304, Sept. 2018.
6. **Z. Chen** and J. Wang, "Underground Odor Source Localization Based on a Variation of Lower Organism Search Behavior," *IEEE Sensors J.*, vol. 17, no. 18, pp. 5963-5970, Jul. 2017.

CONFERENCE

1. **Z. Chen** and N. Xiang, "Further exploration of characteristic impedance measurement at extended frequency range", *J. Acoust. Soc. Am.*, vol. 158, no. 6, pp. A44, 2025 (**invited paper**).
2. **Z. Chen** and N. Xiang, "Wideband characteristic impedance measurement based on cylindrical wave decomposition in an impedance tube," in *Proc. InterNoise 2025*, Sao Paulo, Brazil, 2025.
3. **Z. Chen** and N. Xiang, "Determining the characteristic impedance across a wide frequency range using a three-microphone impedance tube measurement", in *Noise-CON 2025*, Stowe, 2025.
4. **Z. Chen** and N. Xiang, "Uncertainties in parameter estimation of porous media in tube measurements", in *FORUM ACUSITCUM*, Malaga, Spain, 2025. (**invited paper**).
5. **Z. Chen** and N. Xiang, "Characterization of the propagation coefficient and characteristic impedance in an extended frequency range", *ICA2025, J. Acoust. Soc. Am.*, vol. 157, no. 4, pp. A383.
6. M. Ripley, **Z. Chen**, and N. Xiang, "Investigation of errors and uncertainty in low-frequency impedance tube measurements," *ICA2025, J. Acoust. Soc. Am.*, vol. 157, no. 4, pp. A383.
7. C. Gaines, **Z. Chen**, and N. Xiang, "Referenced measurements of diffuser reflections using an acoustic goniometer," *ICA2025, J. Acoust. Soc. Am.*, vol. 157, no. 4, pp. A167.
8. **Z. Chen** and N. Xiang, "Two levels of Bayesian Dissipation Analysis in Impedance Tube Measurements," *J. Acoust. Soc. Am.*, vol. 156, no. 4, pp. A66-A66, 2024.
9. N. Xiang and **Z. Chen**, "A high-resolution acoustic goniometer for characterizing sound diffuser reflections," in *Proc. InterNoise 2024*, Nantes, France, 2024 (**invited paper**).

10. **Z. Chen** and N. Xiang, "Increasing measurement accuracy by estimating multiple dissipations for the impedance tube calibration," in *Proc. InterNoise 2024*, Nantes, France, 2024 (**invited** paper).
11. **Z. Chen** and N. Xiang, "Tube Dissipation estimation for the Transfer function method using Bayesian inference," in *Noise-CON 2024*, New Orleans, 2024.
12. **Z. Chen** and N. Xiang, "Absorptive metasurface inaccuracies made of micropore and microslit panels in multiple structures," *J. Acoust. Soc. Am.*, vol. 155, no. 3, pp. A321, 2024.
13. P. Cunningham, **Z. Chen**, and N. Xiang, "Absorptive performance of layered metastructures using microslit panels," *J. Acoust. Soc. Am.*, 155, no. 3, pp. A340, 2024.
14. **Z. Chen** and N. Xiang, "Dissipation estimation for the impedance tube measurement using Bayesian inference," *J. Acoust. Soc. Am.*, vol. 155, no. 3, 2024 (**invited** paper).
15. N. Xiang and **Z. Chen**, "Experimental characterization of sound diffusers using a portable high-resolution acoustic goniometer," *J. Acoust. Soc. Am.*, vol. 154, no. 4, pp. A44, 2023 (**invited** paper).
16. N. Xiang and **Z. Chen**, "Maximum-entropy principle in knowledge-enhanced machine learning," *J. Acoust. Soc. Am.*, vol. 154, no. 4, pp. A97, 2023 (**invited** paper).
17. **Z. Chen** and N. Xiang, "Uncertainties and Inaccuracies of microslit and microperforated panel absorbers in multiple layers," in *Proc. InterNoise 2023*, Chiba, 2023 (**invited** paper)..
18. **Z. Chen** and N. Xiang, "Boundary conditions determined by experimentally measured surface properties for wave-based time-domain simulations," *J. Acoust. Soc. Am.*, vol. 153, no. 3_supplement, p. A228, 2023.
19. **Z. Chen** and N. Xiang, "Transfer function model-based Bayesian estimation of dissipation and sound speed in impedance tube measurements," *J. Acoust. Soc. Am.*, vol. 152, no. 4, p. A57, 2022.
20. **Z. Chen** and N. Xiang, "Bayesian inference for sound speed and dissipation estimations in impedance tubes," in *Proc. ICA 2022*, Gyeongju, 2022.
21. N. Xiang, **Z. Chen**, J. Zhang, and M. Nolan, "Model-based machine learning in acoustics applications from a Bayesian perspective," in *Proc. ICA 2022*, Gyeongju, 2022 (**invited** paper).
22. **Z. Chen**, C. J. Fackler, and N. Xiang, "Bayesian Parameter Estimation of Microphone Positions, Sound Speed and Dissipation for Impedance Tube Measurements," in *Proc. InterNoise 2022*, Glasgow, 2022.
23. **Z. Chen**, N. Xiang, and K. V. Horoshenkov, "A multipole model-based boundary admittance estimation for wave-based room-acoustic simulations using Bayesian inference," *J. Acoust. Soc. Am.*, vol. 151, no. 4, p. A144, 2021 (**invited** paper).
24. **Z. Chen**, K. V. Horoshenkov, and N. Xiang, "Bayesian inference for boundary admittance estimation using a multipole model for room-acoustic simulation," *J. Acoust. Soc. Am.*, vol. 150, no. 4, p. A348, 2021.

TECHNICAL STRENGTHS

Acoustic Measurement	Impedance tube, Goniometer, Reverberation chamber
Machine Learning	Bayesian inference, Nested sampling
Modeling and Analysis	Matlab, Comsol, FEMM
Circuit Designing	NI multisim, Altium designer
Software & Tools	C/C++, Matlab appdesign, Hardware C, Latex
Misc	DSP, Transducers designing

POSITION & MEMBERSHIP

President	<i>September 2022 - January 2024</i>
ASA (Acoustical Society of America) Student Chapter at RPI	
Treasurer	<i>September 2020 - August 2021</i>
ASA (Acoustical Society of America) Student Chapter at RPI	